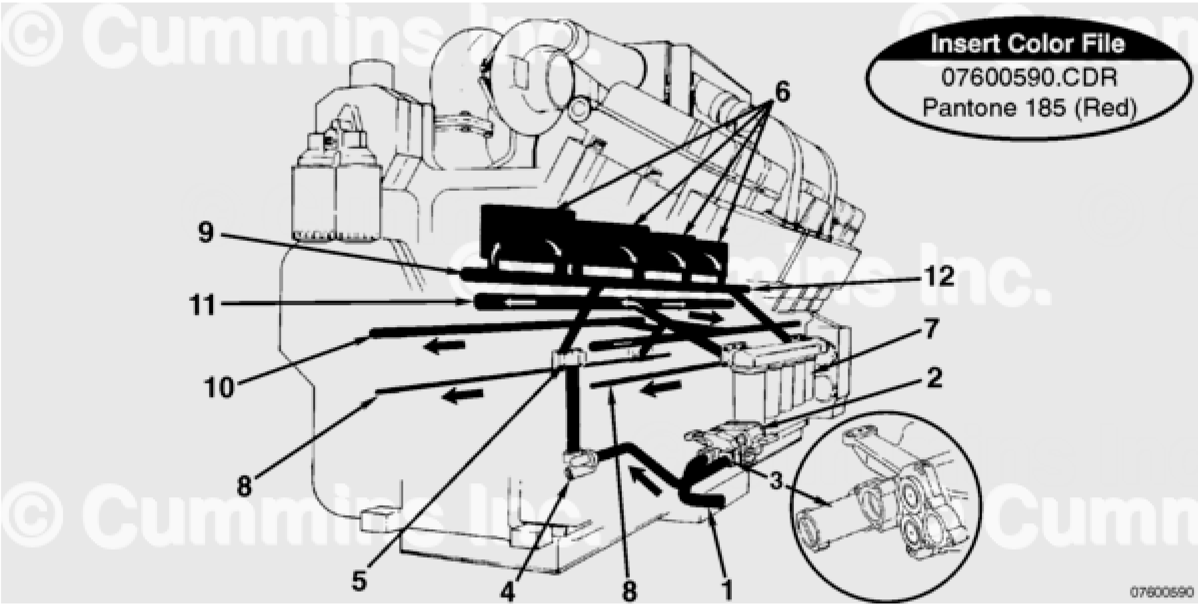
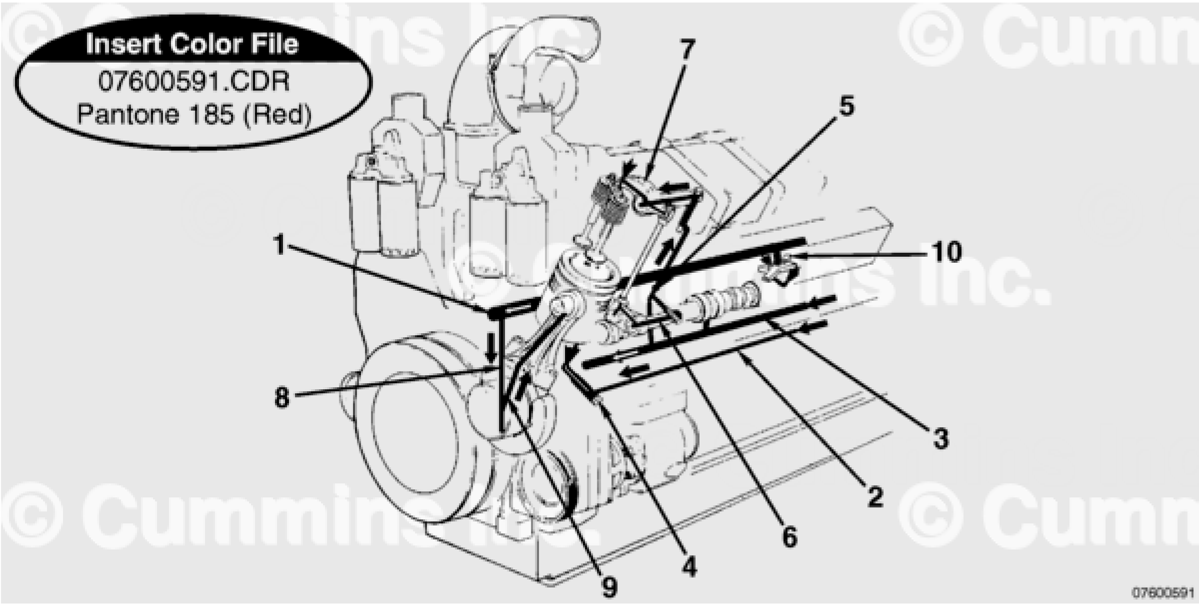
Trainings

Flow Diagram



1. Oil inlet tube
2. Lubricating oil pump
3. High-pressure relief valve (K38 engine **only**)
4. High-pressure relief valve (K50 engine **only**)
5. Jumper cover
6. Oil cooler
7. Oil filter
8. Piston cooling rifle (outboard)
9. Oil rifle/supply to oil coolers
10. Camshaft oil rifle
11. Main oil rifle
12. Cooled oil to filter head.

Note : Older K50 engines were equipped with a high-pressure relief valve attached to the lubricating oil pump front cover.

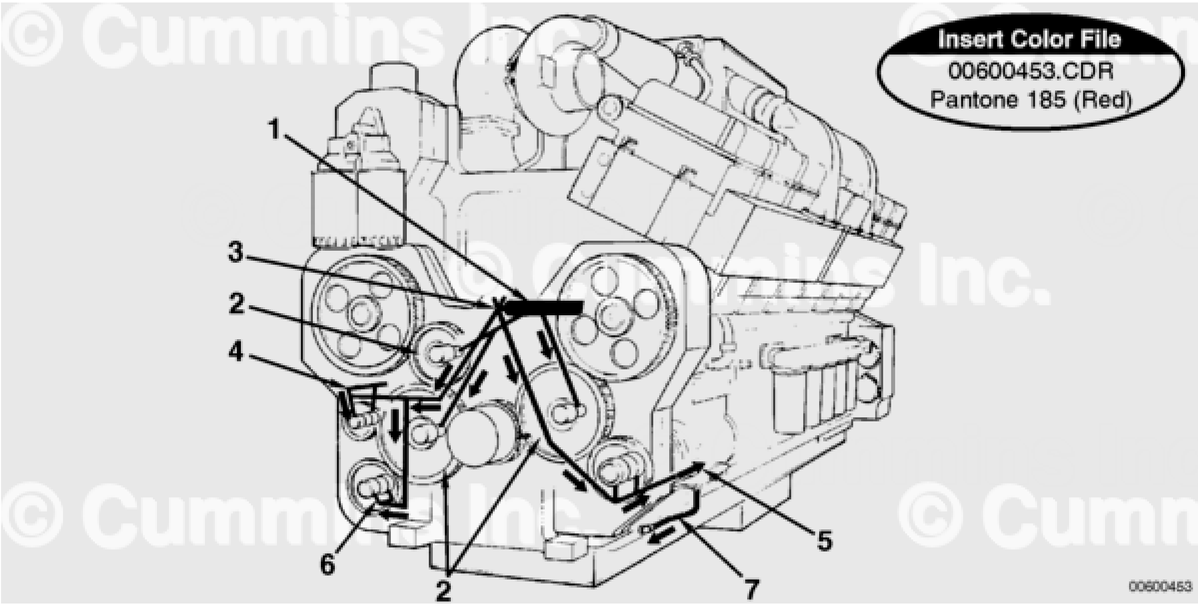


Piston Cooling, Connecting Rod, and Overhead

1. Main oil rifle
2. Piston cooling rifle (outboard)
3. Camshaft oil rifle
4. Piston cooling nozzle (outboard)
5. Orifice
6. Camshaft follower
7. Rocker lever (exhaust)
8. Oil supply to main bearings
9. Oil supply to connection rod
10. Piston cooling nozzle (inboard).

Note : Engines with inboard piston cooling nozzles will not be equipped with items (2) and (4).

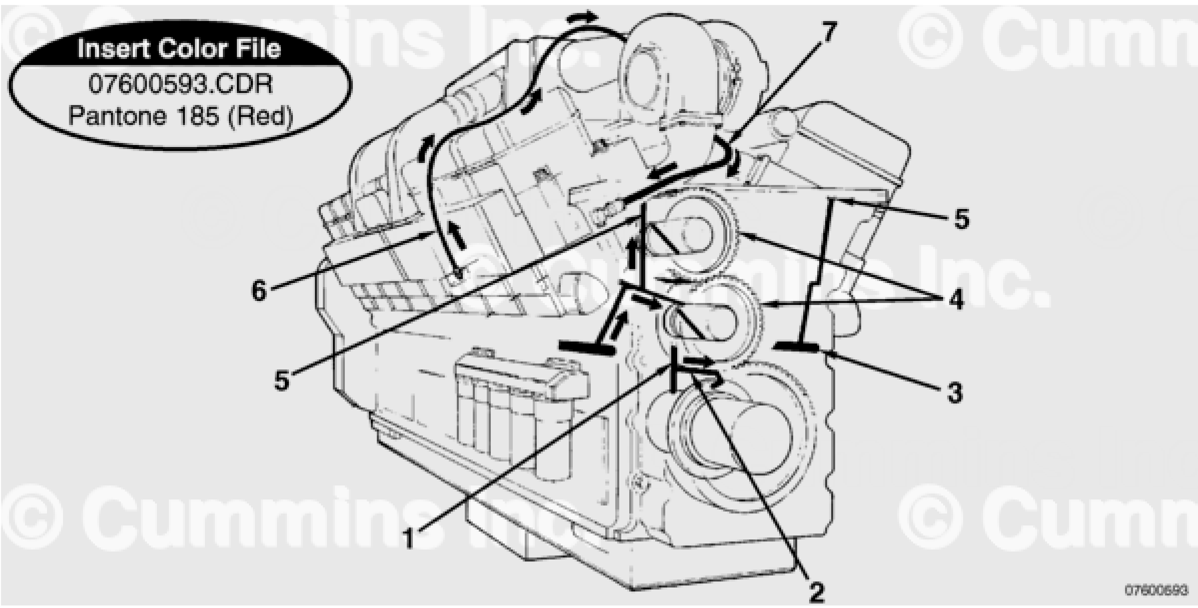
Note : Older engines with outboard piston cooling nozzles will not be equipped with item (10).



Front Gear Train

1. Main oil rifle
2. Idler gear
3. Oil flow through gear housing into front cover
4. Oil to water pump
5. Oil to air compressor
6. Oil to hydraulic pump drive
7. Air compressor oil drain (Cummins® two-cylinder air compressor).

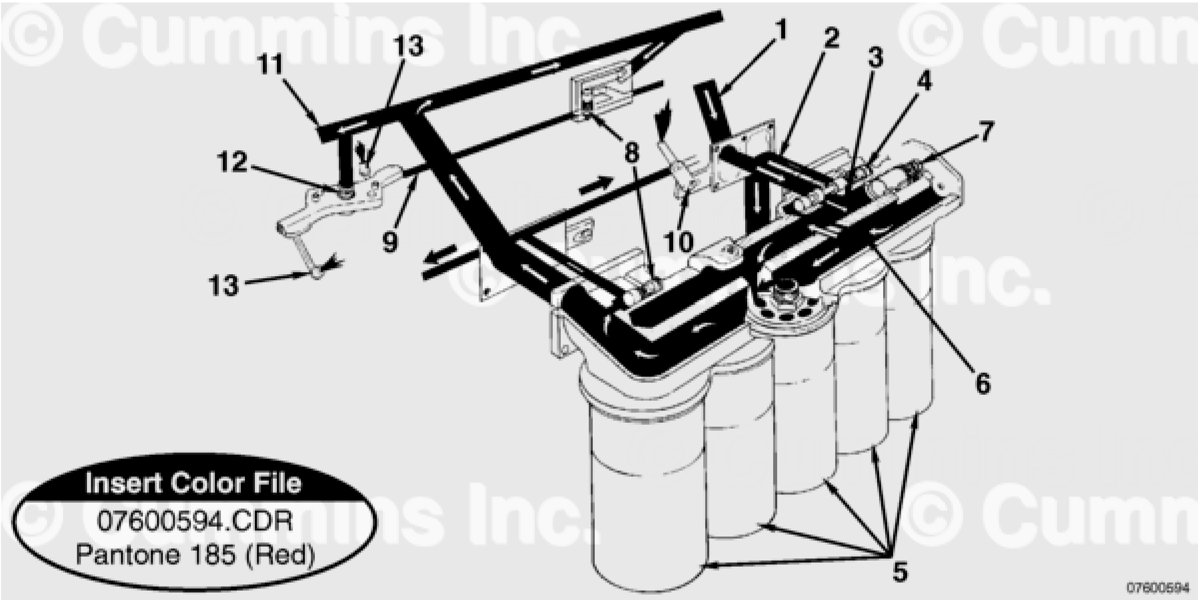
Note : Oil flow to the idler gears (2) is through the cylinder block.



Rear Gear Train, Turbocharger

1. From main oil rifle
2. Oil supply to thrust bearing
3. Camshaft oil rifle
4. Idler gear

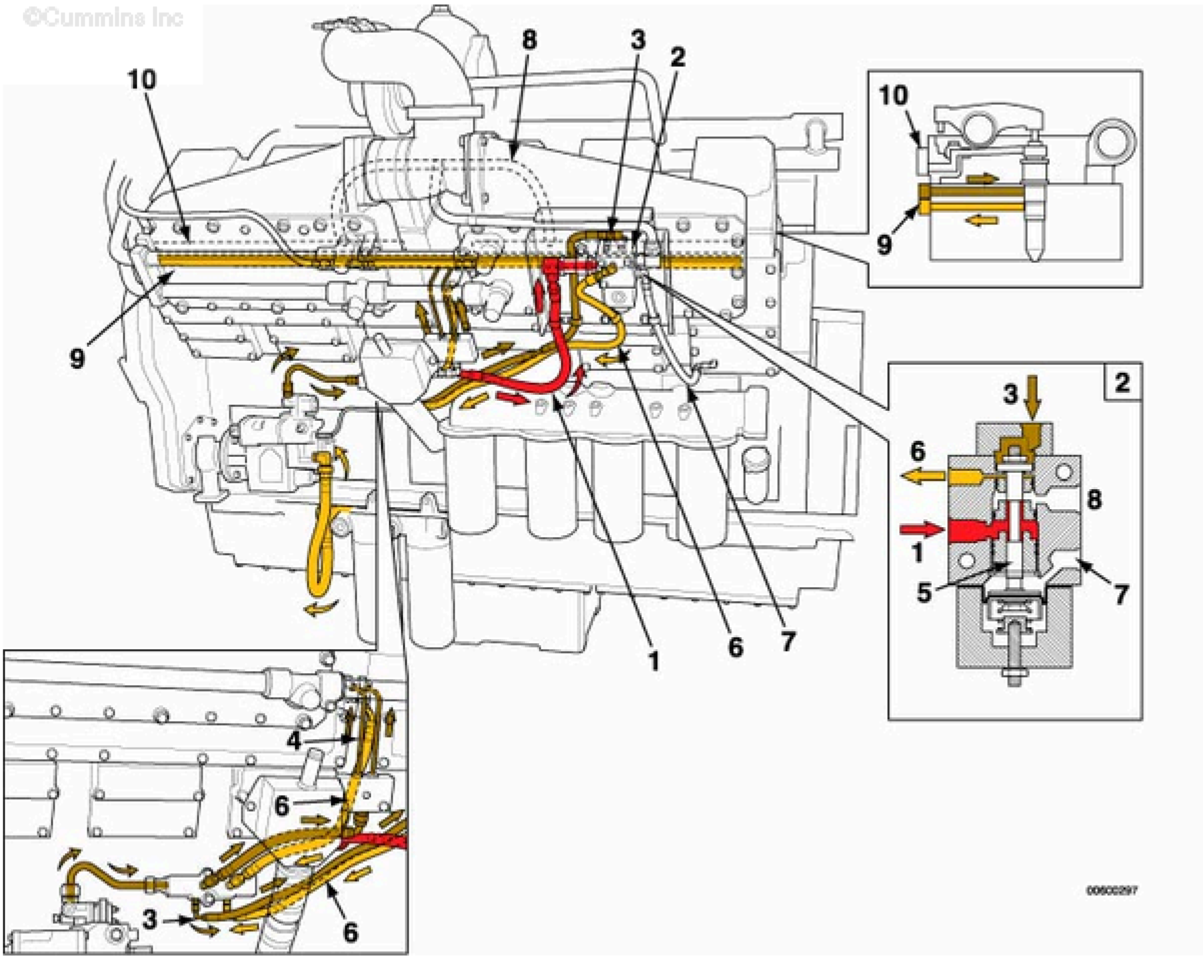
5. Oil supply to upper output housing
6. Turbocharger oil supply
7. Turbocharger oil drain.



Full-Flow Lubricating Oil Filter Head

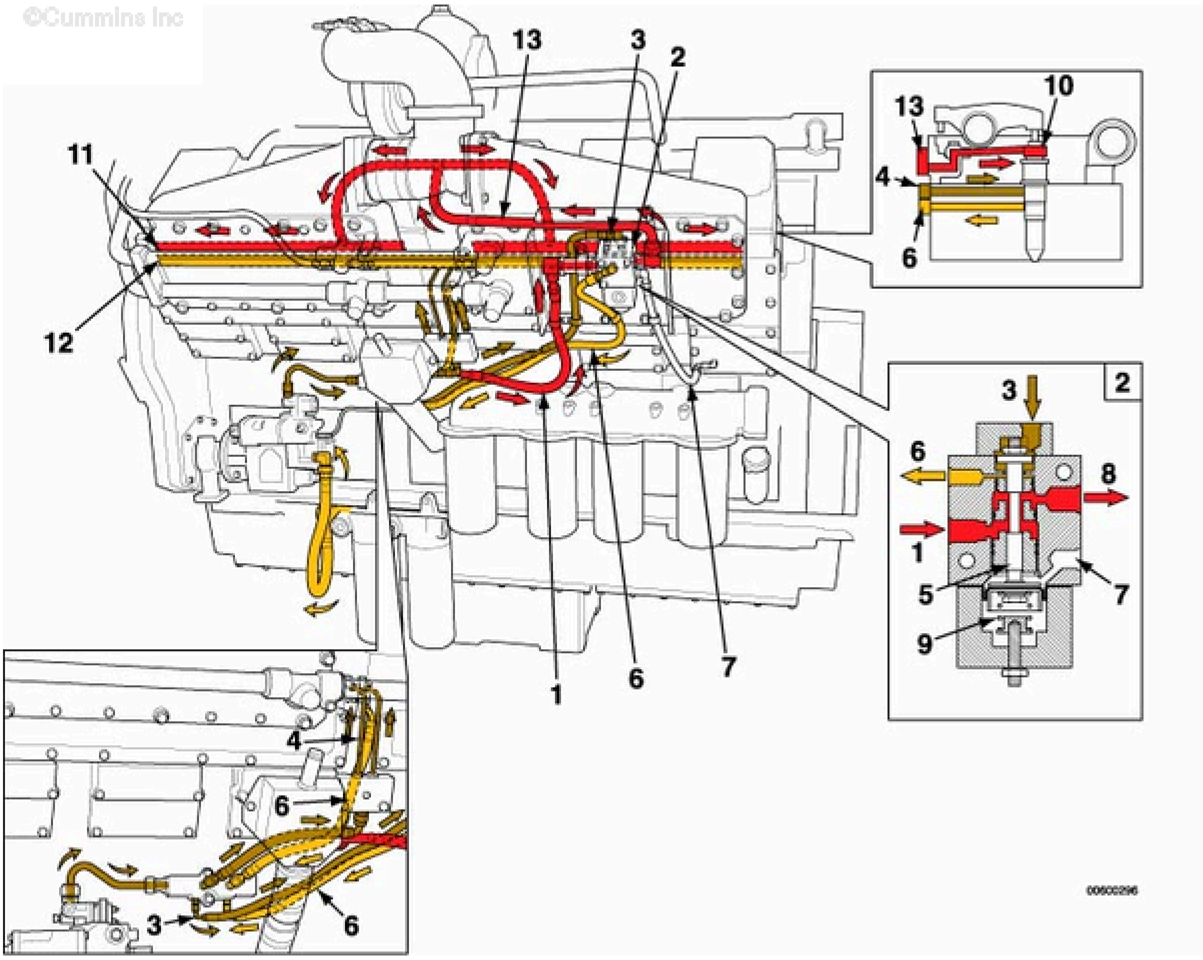
1. Oil supply to filter head
2. Oil return to pan
3. Oil supply to filters
4. Oil pressure regulator
5. Oil filter
6. Control rifle
7. Filter bypass valve
8. Piston cooling control valve (outboard)
9. Piston cooling rifle (outboard)
10. Piston cooling nozzle (outboard)
11. Main oil rifle
12. Piston cooling control valve (center mount)
13. Piston cooling nozzle (center mount).

Note : Engines with center mount piston cooling nozzles will not be equipped with items (8), (9), or (10).



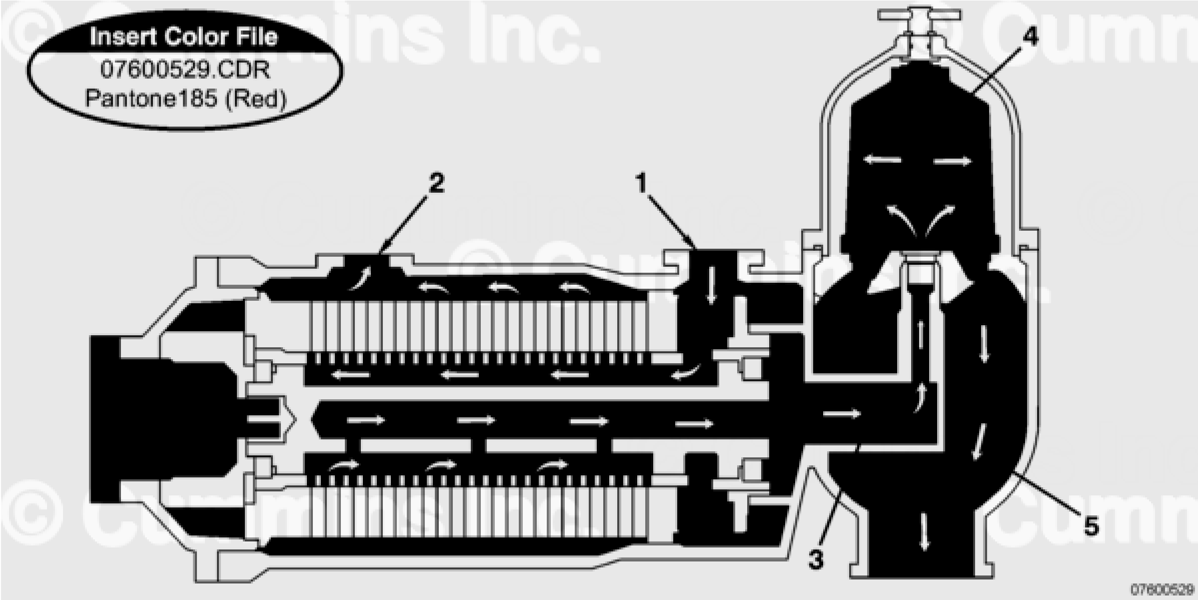
Hydromechanically Controlled STC Lubricating Oil Flow (Normal Timing)

1. Oil from engine rifle (oil rifle pressure)
2. STC control valve
3. Fuel pressure signal to control valve
4. Fuel supply to injectors from pump
5. Oil control plunger
6. Fuel return
7. Vent to camshaft cover
8. Oil supply to tappets (no oil)
9. Fuel rail
10. Oil rail manifold.



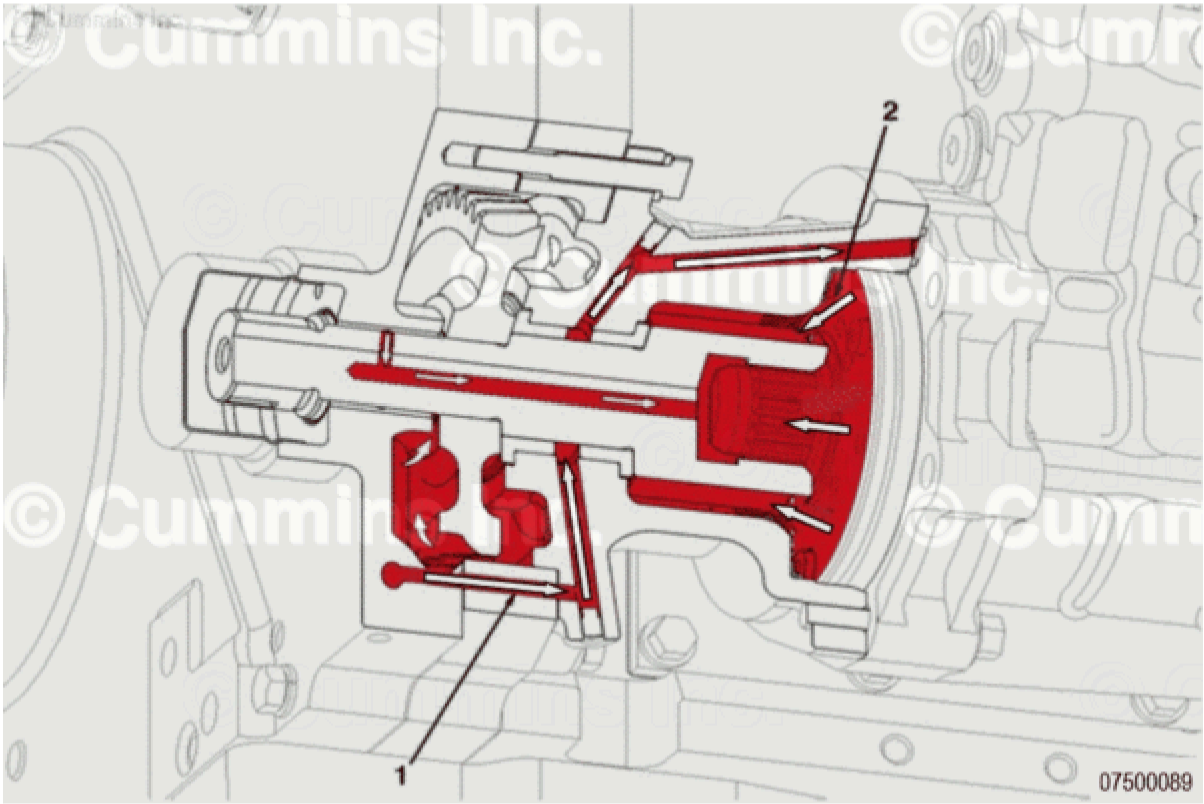
Hydromechanically Controlled STC Lubricating Oil Flow (Advanced Timing)

1. Oil from engine rifle (oil rifle pressure)
2. STC control valve
3. Fuel pressure signal to control valve (fuel pressure)
4. Fuel supply to injectors from pump
5. Oil control plunger
6. Fuel return
7. Vent to camshaft cover
8. Oil supply to tappet
9. Spring
10. STC tappet on injector
11. Oil rail manifold
12. Fuel rail
13. Oil supply to tappets (engine oil rifle pressure).



Eliminator Oil Filter

1. Full-flow contaminated oil to filter
2. Full-flow filtered oil into engine
3. Backflushed oil flow into centrifuge
4. Centrifuge
5. Centrifuged oil back to sump.



Fuel Pump Lubrication - With Electronically Actuated Injectors

1. Lubricating oil supply to accessory drive bushing and fuel pump
2. Lubricating oil return.

Last Modified: 16-Jul-2020